

Rossby Waves on a Beta-Plane

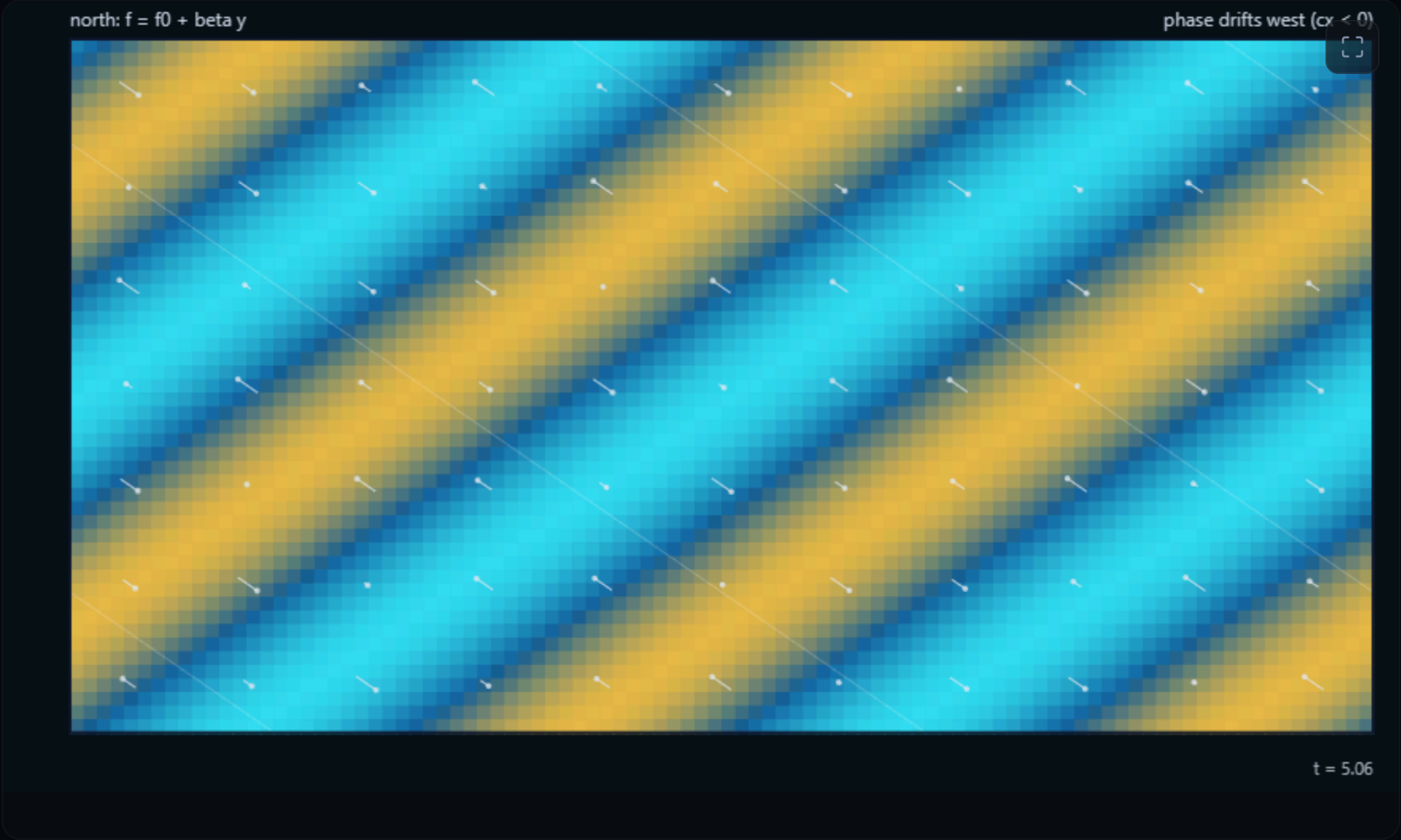
Linear planetary wave $\psi = A \cos(kx+ly-\omega t)$ with $\omega = -\beta k/(k^2+l^2+Rd^{-2})$: streamfunction bands and geostrophic arrows show westward phase propagation.

Pause

Reset phase

On a rotating planet the Coriolis parameter grows poleward. A north-south displacement changes planetary vorticity, so the restoring motion forms large-scale Rossby waves with westward phase speed.

LIVE GRAPHS



ROSSBY MODE

Beta gradient β

1.4

Zonal wavelength λ_x

8

Meridional wavelength λ_y

5.5

Deformation radius Rd

12

ANIMATION

Streamfunction amplitude

1

Time speed

0.75 x

The color field is streamfunction ψ . Arrows show geostrophic velocity $(u, v) = (-\psi_y, \psi_x)$. The pattern moves west because β makes ω/k negative for $k > 0$.

MEASURED VALUES

ω	-0.570
phase speed c_x	-0.726
group speed c_{gx}	-0.262
period	11.02

About this model

Rossby waves are large-scale planetary waves produced by the meridional gradient of the Coriolis parameter, $f = f_0 + \beta y$. This simulator uses a single linear beta-plane mode with streamfunction $\psi = A \cos(kx + ly - \omega t)$. The dispersion relation $\omega = -\beta k / (k^2 + l^2 + Rd^{-2})$ is evaluated directly, so no numerical PDE stability limit is hidden in the animation. For $\beta > 0$ and eastward wavenumber $k > 0$, the zonal phase speed $c_x = \omega/k$ is negative: crests drift west even though the group velocity can change sign depending on the aspect ratio and deformation radius.

Who it's for: Physical oceanography, atmospheric dynamics, or geophysical fluid dynamics introductions.

KEY TERMS

- Rossby wave
- Beta-plane
- Planetary vorticity
- Geostrophic flow
- Dispersion relation

How it works

Linear barotropic / shallow-water Rossby wave on a beta-plane. Tune β , wavelengths, and deformation radius to see why planetary vorticity gradients give westward phase propagation.

Key equations

$$\psi = A \cos(kx + ly - \omega t), \quad k = 2\pi/\lambda_x, \quad l = 2\pi/\lambda_y$$

$$\omega = -\beta k / (k^2 + l^2 + Rd^{-2}), \quad c_x = \omega/k < 0 \text{ for } \beta > 0$$

Frequently asked questions

Is this a full shallow-water forecast model?

No. It is one analytic linear normal mode. That keeps the key beta-effect and dispersion relation visible, but excludes forcing, dissipation, boundaries, topography, and nonlinear wave-mean-flow interactions.

Why can group velocity differ from phase velocity?

Rossby waves are dispersive: ω is not proportional to k . Crests move with $c_x = \omega/k$, while wave-packet energy follows $\partial\omega/\partial k$, which can be eastward or westward depending on k , l , and Rd .

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Gap d , cylinders Ω_i , Ω_o , kinematic v : Taylor number Ta and critical Ta_c for toroidal rolls — cartoon + formula.